

Mirror Descent Examples

Motivating example: $\min: \|Ax - b\|_1$ this is L -Lipschitz w.r.t. $\|\cdot\|_1$
but incurred $\sqrt{n}$ factor to convert to $\|\cdot\|_2$.

$$\min: f(x)$$

$$\text{st: } x \in \Delta_n,$$

$$\Delta_n = \{x \in \mathbb{R}^n : x \geq 0, \sum x_i = 1\}$$



Assume: f 1-Lipschitz w.r.t. $\|\cdot\|_1$
equivalently: $\|\nabla f(x)\|_\infty \leq 1$.

Recall: $\|\nabla f(x)\|_2 \leq \sqrt{n}$ Therefore: sub-grad descent: $\frac{\|x_0 - x^*\|_2 \cdot L}{\sqrt{T}}$
 $= \|x_0 - x^*\|_2 \cdot \left(\sqrt{\frac{n}{T}}\right)$

MD: $x_{t+1} = \arg\min_{x \in \mathcal{X}} \eta g_t^T x + D_\phi(x, x_t)$

What function ϕ to use?

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$$\phi(x) = \sum x_i \log x_i$$

Fact: ϕ is 1-strongly convex on Δ w.r.t. $\|\cdot\|_1$
 $\Rightarrow \underline{\rho = 1.}$

$$R^2 = D_\phi(x_1, x^*) \quad \text{if } x_1 = \begin{pmatrix} 1/n \\ 1/n \\ \vdots \\ 1/n \end{pmatrix}$$
$$\leq \log n$$

Plugging in to MD result:

$$f\left(\frac{1}{T} \sum_{t=1}^T x_t\right) - f(x^*) \leq R L \sqrt{\frac{2}{\rho T}} = \sqrt{2} \cdot \sqrt{\frac{\log n}{T}}$$

Compare to sub-gr. desc: $\sqrt{\frac{n}{T}}$.

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What is MD doing?

proj sub-gr. desc: $x_{t+1} = \text{Proj}_\Delta (x_t - \eta g_t)$.

MD: $x_{t+1} = \argmin_{x \in \Delta} : \eta g_t^T x + D_\phi(x, x_t)$

$$\boxed{0 \in \eta g_t + \nabla \phi(x_{t+1}) - \nabla \phi(x_t)} + N_{\mathcal{X}}(x_{t+1})$$

Can show: x_{t+1} is the Bregman projection of y_{t+1} onto $\mathcal{X} = \Delta$
i.e., $x_{t+1} = \argmin_{x \in \Delta} : D_\phi(x, y_{t+1})$

y_{t+1} is given by: $\nabla \phi(y_{t+1}) = \nabla \phi(x_t) - \eta g_t$.

For $\phi(x) = \sum x_i \log x_i$, can show:

$$y_{t+1}(i) = x_t(i) \cdot \exp\{-\eta (\nabla f(x_t))_i\} \leftarrow \text{compute this}$$

$$x_{t+1} = \frac{y_{t+1}}{\|y_{t+1}\|_1}.$$



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